Check for updates

Host specificity of the gut microbiome

Elizabeth K. Mallott[1,2] *and Katherine R. Amato*[1]✉

Abstract | Developing general principles of host–microorganism interactions necessitates a robust understanding of the eco-evolutionary processes that structure microbiota. Phylosymbiosis, or patterns of microbiome composition that can be predicted by host phylogeny, is a unique framework for interrogating these processes. Identifying the contexts in which phylosymbiosis does and does not occur facilitates an evaluation of the relative importance of different ecological processes in shaping the microbial community. In this Review, we summarize the prevalence of phylosymbiosis across the animal kingdom on the basis of the current literature and explore the microbial community assembly processes and related host traits that contribute to phylosymbiosis. We find that phylosymbiosis is less prevalent in taxonomically richer microbiomes and hypothesize that this pattern is a result of increased stochasticity in the assembly of complex microbial communities. We also note that despite hosting rich microbiomes, mammals commonly exhibit phylosymbiosis. We hypothesize that this pattern is a result of a unique combination of mammalian traits, including viviparous birth, lactation and the co-evolution of haemochorial placentas and the eutherian immune system, which compound to ensure deterministic microbial community assembly. Examining both the individual and the combined importance of these traits in driving phylosymbiosis provides a new framework for research in this area moving forward.

Phylosymbiosis
Similarities in host-associated microbial community structure that mirror the phylogenetic relationships of the hosts.

Vertical transmission
A pathway of transmission where a symbiotic organism is transmitted from the host parent to offspring.

Co-evolution
Reciprocal genetic change in two species that occurs as a result of the selective pressures that each imposes on the other.

[1]*Department of Anthropology, Northwestern University, Evanston, IL, USA.*

[2]*Department of Biological Sciences, Vanderbilt University, Nashville, TN, USA.*

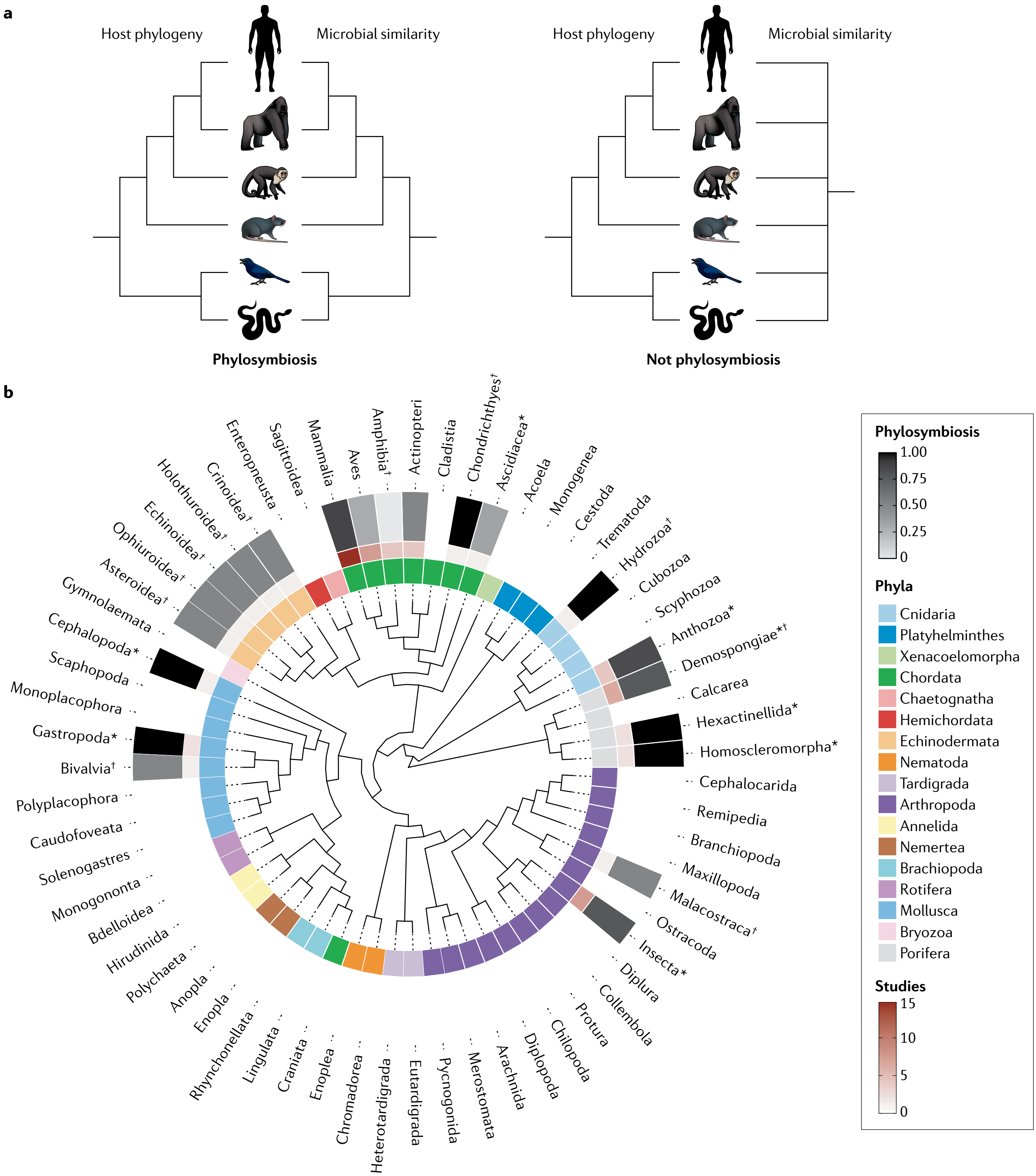
✉*e-mail: katherine.amato@northwestern.edu*

https://doi.org/10.1038/s41579-021-00562-3

As our understanding of the impact of the gut microbiota on host metabolism, immune function and behaviour has expanded[1–4], there has been widespread interest in determining the eco-evolutionary processes that structure microbiomes. To this end, foundational animal microbiome research examined the extent to which gut microbiome composition and function are correlated with host ecological traits, such as dietary niche versus host phylogenetic background[5,6]. This work, and subsequent studies using other analyses and animal systems, indicated that host dietary niche has a strong impact on the gut microbiome and can lead to the convergence of gut microbiomes of host species with distinct phylogenetic backgrounds[7–9]. The implication that the microbiome is not completely constrained by host genetics has led to important work considering the contribution of the microbiota to host ecological plasticity[10–12]. However, with increasing research, it has also become apparent that the effect of host dietary niche on the gut microbiome does not preclude the effect of host phylogeny. Even in studies that detect an effect of host dietary niche on the microbiome, the effect of host phylogeny is often greater[5,7,13]. These patterns indicate that the dichotomy of host dietary niche versus phylogeny, or host ecology versus evolution, is false. Instead, host phylogeny influences host adaptations to specific ecologies, which simultaneously shape the microbiome[13–15].

Observed patterns of phylosymbiosis or "microbial community relationships that recapitulate the phylogeny of their host"[16] (FIG. 1), can provide crucial insight into microbiota–host interactions. First, given the ability of microbiota to influence host phenotypes, there is growing interest in testing the extent to which these dynamics can shape patterns of host ecology and evolution. However, for selection to act on microbially influenced host phenotypes, those phenotypes must be heritable. This requires host species-specific microbiota traits that are reliably associated with hosts across generations, regardless of whether they are acquired or not acquired through vertical transmission. Although phylosymbiosis is not the only scenario in which this dynamic occurs, it is a clear example of it. As a result, observed patterns of phylosymbiosis have fuelled a growing interest in causally linking the microbiome to host evolutionary trajectories[12,17,18].

Additionally, phylosymbiosis is a powerful framework for informing our understanding of host-associated microbial community assembly. Whereas individual microbial lineages commonly co-diversify with specific host species, phylosymbiosis occurs when the composition of the overall microbial community, or a subset of multiple lineages, can be predicted by host phylogeny. Although some microbial lineages may still co-diversify with hosts, phylosymbiosis itself is not an indicator of host–microorganism co-divergence or co-evolution[19]. Other ecological processes, including dispersal to hosts and priority effects, selection of microbial lineages by the host environment, and ecological drift, can also shape the



microbial community and patterns of phylosymbiosis[20]. Identifying the contexts in which phylosymbiosis does and does not occur, as well as the factors that strengthen or weaken it in any given system, can allow a more robust evaluation of the relative importance of these processes in shaping the microbial community. Identifying smaller subsets of the overall microbial community in which phylosymbiosis is stronger or weaker can also serve a similar function.

In this Review, we summarize microbial community assembly patterns contributing to phylosymbiosis. We then integrate existing data describing phylosymbiosis across animal systems with the goal of better identifying the contexts in which it occurs and the processes that drive it. We conclude that phylosymbiosis, as measured in the microbiome as a whole, becomes less prevalent as microbiomes become taxonomically richer across the animal kingdom. We hypothesize that this pattern

◀ Fig. 1 | **Current evidence of phylosymbiosis in animals. a** | A schematic depicting the concordance between host phylogeny and microbiome similarity that indicates phylosymbiosis and the lack of concordance with host phylogeny when the microbiota is stochastically assembled. **b** | A phylogenetic tree of classes within Animalia with phyla annotated on the inner ring, the number of studies examining phylosymbiosis in each class annotated on the middle ring and the proportion of studies finding evidence for phylosymbiosis in host-associated animal microbiomes annotated on the outer ring. Darker red indicates that more published studies have examined whether phylosymbiosis is present or not present in a given class. Classes of animals where phylosymbiosis is commonly observed are annotated in a darker grey than classes where evidence is mixed or suggests phylosymbiosis does not occur, and classes of animals for which there is no evidence for or against phylosymbiosis are annotated in white. For many classes, data are drawn from skin (dagger) or other non-gut tissues (asterisk) as data from the gut microbiome cannot be collected. The phylogenetic tree is based on data from TimeTree; the tree was visualized in R with use of the ggtree package and annotations were added on the basis of the references provided in Supplementary Table 1 and the analysis presented in Supplementary Table 2. Briefly, phylosymbiosis prevalence was calculated as the proportion of studies showing support for phylosymbiosis within a given class. Studies with mixed results were treated as partial support — if three sixths of metrics within a study showed support for phylosymbiosis, that study was weighted by 0.5.

emerges in complex microbial communities because community assembly processes are less likely to uniformly influence individual components of a complex community. Interestingly, mammals appear to be an exception to this pattern. Despite having taxonomically rich microbiomes, mammals commonly exhibit phylosymbiosis. We posit that this exception is a result of a unique combination of mammalian traits that compound to strengthen the influence of multiple individual community assembly processes. Examining the relative importance of these host traits and community assembly processes in driving phylosymbiosis provides a new framework for research in this area moving forward.

Drivers of phylosymbiosis

General patterns of host-associated microbial community assembly, including the special case of phylosymbiosis, are driven by a variety of host–microorganism and microorganism–microorganism interactions (FIG. 2). Underlying these interactions are the ecological community assembly processes of dispersal, selection, competition, diversification and drift[21]. However, although these terms are useful for defining the mechanistic pathways by which the microbiota assembles itself, they do not describe what proportion of the microbial community is likely to be structured by these processes. Some processes are more likely to affect individual microbial lineages, whereas others affect the entire community. Additionally, these terms do not distinguish between the host and microbial traits that can independently contribute to each process. For example, both hosts and microorganisms can affect patterns of microbial dispersal in distinct ways. The importance of these processes, as well as host versus microbial contributions to them, is likely to differ depending on the host system and the environment[15,19,21–23]. Therefore, identifying the specific host and microbial contributions to different processes of microbial community assembly will allow us to develop more generalizable rules for host–microbiota associations and provide key insight into the prevalence and strength of phylosymbiosis across a range of host taxa.

Although the microbiomes of multiple animal body sites have been tested for phylosymbiosis[20,24–27], the gut microbiome is the most well studied. However, overall, the same processes of microbial community assembly that lead to phylosymbiosis in the gut are likely to lead to phylosymbiosis at other body sites, albeit with potential differences in the relative importance of each process and the strength of phylosymbiosis[15]. Because of these similarities, we discuss the processes driving phylosymbiosis generally and do not specify body sites. Specifically, we discuss host effects on the processes of microbial dispersal, selection and diversification. We also discuss microbial effects on the processes of microbial dispersal and selection as well as host diversification.

Host influences on microbial dispersal. Hosts can influence microbial dispersal via both vertical transmission and horizontal transmission (FIG. 2). In some animals, vertical transmission occurs because intracellular symbiotic microorganisms colonize the germ line[28–33]. However, this pathway of microbial community assembly is physically constrained to the transmission of a select few microbial lineages. Vertical transmission of a larger number of microbial lineages, or the microbiota as a whole, can also occur through other physiological and behavioural pathways in both invertebrates and vertebrates. In oviparous animals, eggs are inoculated via contact with microorganisms on ovipositors or in maternal secretions[32,34,35], and in viviparous animals, infant contact with maternal microbiota during birth provides opportunities for microbial transmission[36,37]. In animals with parental care behaviours, prolonged periods of physical contact between parents and offspring also encourage microbial transfer[38]. For example, incubation of eggs by birds regulates egg pathogen loads[39], and saliva exchange between parents and nestlings during feeding alters the nestling microbiome[40]. Similarly, *Nicrophorus* burying beetle offspring, as well as the offspring of some stinkbug families, are colonized by microorganisms from their parents as a result of multiple prehatching and posthatching behaviours[41,42]. In other insects, eggs are exposed to microorganisms in the reproductive organs or through microbial seeding once eggs are laid[28,34,43].

Hosts can also affect patterns of microbial dispersal via horizontal transmission of potentially large portions of the microbiota as a result of host interactions with conspecifics and the surrounding environment. Within host species, close physical proximity and frequent social interactions lead to convergence in microbiomes as a result of horizontal transmission[44,45]. Similar dynamics appear to operate across host species[46]. Host biogeography and behaviour can also influence which microorganisms hosts encounter in their environment[21,23,47]. For example, differences in host diet and substrates lead to distinct gut microbial communities in *Drosophila* species[48]. Host geography structures the gut microbiomes of mammals in the Americas independently of diet or phylogeny[47], likely due in part to variation in which bacteria are available to be acquired from the environment.

Priority effects
A phenomenon in which species that arrive first at a site alter the abiotic and/or biotic conditions such that this impacts the colonization success of species arriving later.

Ecological drift
The random fluctuation of species abundances in a community or population, often as a result of isolation from other populations.

Horizontal transmission
A pathway of transmission where a symbiotic organism is transmitted between individuals via the social or physical environment instead of from parent to offspring.

Oviparous animals
Animals that lay eggs and in which embryonic development does not occur internally in the mother.

Ovipositors
Tubes through which a female animal lays or deposits eggs; most often refers to a structure found in insects.

Viviparous animals
Animals characterized by live birth (viviparity) of offspring with embryonic development occurring internally in the mother.

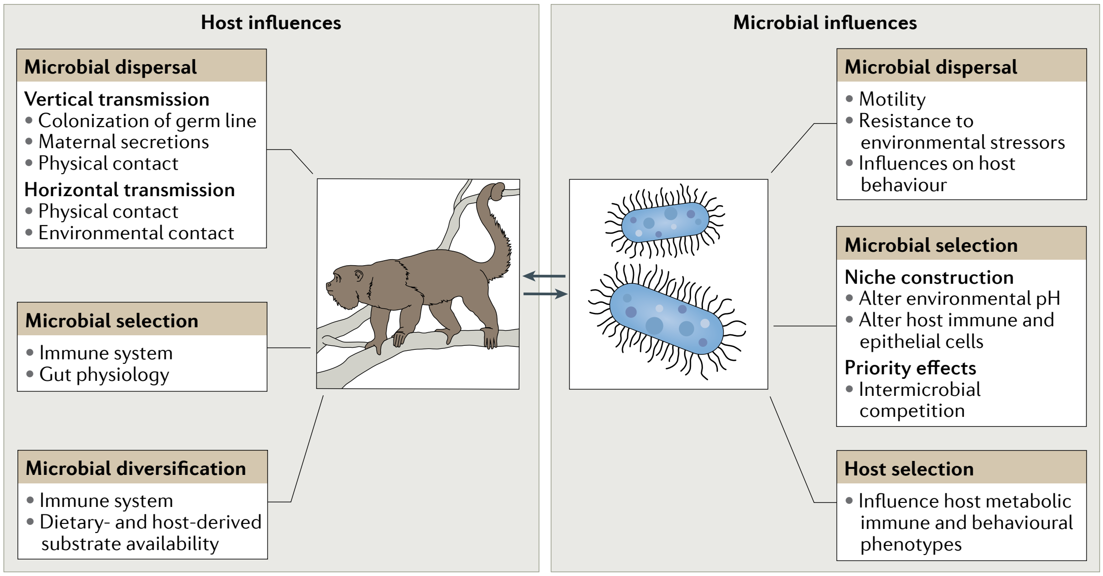


Fig. 2 | **Drivers of microbial community assembly and phylosymbiosis.** Phylosymbiosis is driven by both host and microbial influences on microbial community assembly processes. Hosts affect microbial dispersal via vertical and horizontal transmission pathways. They also use physiological filters to select which microorganisms colonize successfully. Hosts can exert selective pressure on colonizing microorganisms that results in microbial diversification. Microbial traits also affect patterns of dispersal by affecting the likelihood of contact with hosts. Microorganisms can influence the selective forces determining which microorganisms successfully colonize hosts once they are exposed. Some microbial lineages engage in niche construction in which they alter the local host environment to increase the probability of survival and decrease competition with other microorganisms. Priority effects in which microorganisms that disperse to hosts first outcompete those that disperse later also contribute to this process. Microorganisms can contribute to host diversification, which can feed back and affect the other community assembly processes outlined, by altering host phenotypes in ways that affect host fitness.

Host influences on microbial selection. Once hosts are exposed to microorganisms through either vertical transmission or horizontal transmission, host filtering allows some microorganisms to stably colonize, while others are excluded, acting as an important mechanism shaping overall microbial community structure[15] (FIG. 2). Host filters can include both environmental conditions in the gut that preclude survival of some microbial populations and promote the growth of others, and direct host–microorganism signalling, which ranges from microbial lineage specific to more general[49–51]. In most animal systems, we know very little about these specific host filtering mechanisms. However, the innate immune system has a role in filtering via microbe-associated molecular patterns that can exclude a range of microbial lineages[52]. Additionally, in vertebrates, the adaptive immune system generally has a key role in filtering the microbiota[53–55]. Even though invertebrates and vertebrates have similar microbial exposure, there is a higher microbial diversity stably associated with vertebrate hosts compared with invertebrate hosts, and it has been suggested that a key driver of the evolution of the adaptive immune system in vertebrates, particularly warm-blooded vertebrates, was the maintenance of a higher diversity of microbial symbionts[56–58]. A recent analysis demonstrating increased microbial diversity associated with increased adaptive immune system complexity in vertebrates provides additional support for this hypothesis[59]. Specific immune mutations also appear to affect the gut microbiota substantially. For example, in mice, genetic mutations that eliminate mature lymphocytes result in increased relative abundances of microbial taxa such as *Akkermansia*[60].

Additionally, other aspects of host physiology, such as gut anatomy and function, may also act as filters for large proportions of the microbiota. For example, the size of the gut as well as the transit time and patterns of peristalsis can all affect the microbial lineages that persist in the gut[61–63]. Where fermentation happens in the gut (foregut versus hindgut), the diversity and abundance of host digestive enzymes in each gut section can also affect the chemical environment of the gut and act as a filter for microorganisms[64,65]. For example, the presence or absence of a sacculated foregut appears to have a strong influence on the composition and function of the primate gut microbiome[13–15]. Similarly, experimental selection of voles towards a high-fibre, herbivorous diet results in distinct microbial communities that may be driven by selective pressure on the microbiota exerted through changes in gut physiology[66].

Host influences on microbial diversification. Once a microorganism is established in a host, host physiology can impose selective pressures on it that result in within-host microbial evolution[18] (FIG. 2). This process often involves individual microbial lineages and can occur via multiple mechanisms. In vertebrates, for example, the adaptive immune system can both hasten the pace of microbial evolution and act as a constraining factor[67,68]. The availability of dietary substrates can also

Innate immune system
A non-specific immune response to potential pathogens found in invertebrates and vertebrates. In vertebrates, the innate immune system has a role in activating the adaptive immune system.

Adaptive immune system
The portion of the immune system that, in vertebrates, allows an organism to identify, learn about and respond to specific pathogens.

Sacculated foregut
A multichambered stomach which allows microbial fermentation before food moves further into the digestive tract.

exert selective pressure on microbial metabolism as the genetic pathways necessary to break down host-derived or diet-derived glycans are distinct[69]. For example, cell envelope biosynthesis genes appear to be under selection in human-associated *Bacteroides fragilis*, potentially to reduce immune detection or increase utilization of host-derived carbohydrates[70].

Existing data suggest that within-host microbial evolution is most often a result of microbial adaptation within a host's lifetime and does not commonly occur across host generations[71]. For example, *B. fragilis* exhibits unique diversity in cell envelope biosynthesis and polysaccharide utilization genes among host individuals[70,71]. Nevertheless, high rates of vertical and/or horizontal transmission can allow within-host evolution across generations[67,68,72]. As a result, many examples of intergenerational within-host evolution involve microbial lineages that colonize the germ line (for example, *Buchnera* species in aphids)[73]. However, within-host evolution of microbial lineages may also occur across host generations in non-endosymbiotic microorganisms[71]. For example, Bifidobacteriaceae and Bacteroidaceae lineages exhibit similar divergence times as their hosts, potentially as a result of microbial co-diversification with hosts[74,75]. Nevertheless, data are limited, and it is not yet well understood when we would expect within-host microbial evolution of a small number of microbial lineages to lead to a permanent shift in overall community structure or function[71].

Microbial influences on microbial dispersal. Microbial traits can affect the likelihood of microbial dispersal to hosts (FIG. 2). However, unlike most host influences on microbial dispersal, many microbial traits that influence dispersal dictate interactions between a single microbial lineage and potential hosts. For example, the infant gut is initially colonized by a small number of fast-growing, oxygen-tolerant, flagellated bacteria that can survive in harsh environments[76]. Likewise, the evolution of *Aeromonus veronii* in a zebrafish system demonstrates that improved microbial motility allows more rapid host entry from the environment and increases the probability of colonization[77]. By contrast, some microbial taxa influence host behaviour[3], which has the potential to affect dispersal of larger portions of the microbiota. For instance, if specific behaviours such as parental care or patterns of social interaction with conspecific networks are affected, it could improve the probability of colonization by complex microbial populations within and across host generations[78,79].

Microbial effects on microbial selection. Microorganisms can affect selective regimes in the host through interactions with both hosts and other microorganisms (FIG. 2). First, microorganisms can engage in niche construction by influencing host physiology in ways that increase the likelihood and strength of their associations with hosts. Microorganisms can affect the structure of the host epithelial cell layer and the turnover of epithelial cells[80], which impacts the physical niches that microorganisms can inhabit as well as the host energy substrates to which they have access. Microorganisms also programme host immune function and can facilitate host tolerance by silencing Toll-like receptor 4 (TLR4) signaling[81] and altering the function of immune cells, including dendritic cells and a range of T cells (regulatory T (T_{reg}) cells, T helper cells and natural killer cells)[82]. As a result, microbial taxa that can alter the host gut environment may be more likely to associate with hosts, independently of host traits designed to filter colonization by specific microbial taxa. This process is potentially relevant to individual microbial lineages as well as key subsets of the overall community.

Additionally, intermicrobial competition can directly affect which microorganisms successfully colonize the host. Once exposed to hosts, some microbial taxa secrete compounds that alter environmental conditions to increase their survival and negatively impact the survival of other microorganisms[52]. For example, short-chain fatty acid production by bacteria that ferment carbohydrates decreases the pH of the gut and the vaginal tract[83], precluding acid-sensitive microorganisms from becoming established[84]. Microorganisms that disperse to hosts earlier in the colonization process can also exclude others that arrive subsequently by stably colonizing available niches and limiting access to resources, an ecological phenomenon known as priority effects. For example, colonization of the infant gut by specific *Bacteroides* strains results in the exclusion of other *Bacteroides* strains subsequently and may trigger specific patterns of microbial community succession[85]. Again, these dynamics can operate between individual microbial lineages or affect larger portions of the microbial community.

Microbial effects on host selection. Microorganisms may also affect host physiology in ways that alter host responses to selective pressure and result in host evolution (FIG. 2). For example, microbial contributions to host nutrition or immune function could alter host survival and reproductive success in certain environments[86,87]. If the presence or absence of these fitness-promoting microorganisms is associated with variation in host genetics, the resulting changes in allele frequencies would promote microbially incited host divergence that could ultimately lead to patterns of host–microorganism co-diversification and phylosymbiosis. This process could affect both individual microbial lineages and the overall microbial community and its emergent functions. Although microbial effects on host evolution have been explored theoretically[17,88], few empirical data have been generated to directly demonstrate this phenomenon. However, some evidence of it does exist. For instance, *Drosophila melanogaster* populations experimentally exposed to distinct microbial communities exhibit distinct allele frequencies in the span of five generations[89]. Additionally, it has been demonstrated that host–microorganism mismatches contribute to lethality in *Nasonia* wasp hybrids and therefore have a role in driving host species reproductive isolation[16].

Evidence of phylosymbiosis

Although phylosymbiosis is commonly assumed to be a widespread phenomenon, a close inspection of the data suggests that may not be the case. Currently, 21 classes

Co-diversification
Diversification of two species at the same pace and that have a shared evolutionary history as a result of either co-evolution or a shared environment.

Niche construction
When a species changes its local environment, altering the selective pressures acting on it and other organisms in its environment.

Succession
The process of change in the composition of an ecological community over time, often involving increasing community diversity.

of animals have been tested for phylosymbiosis in both the laboratory and the wild, and evidence for it differs across studies (and even within studies depending on the metric used) (FIG. 1; Supplementary Table 1). For many classes, such as ascidians (Ascidiaea), bivalves (Bivalvia) or ray-finned fish (Actinopteri), patterns are difficult to interpret given that data are limited to only a handful of studies and/or host taxa (FIG. 1; Supplementary Table 1). However, even for birds and amphibians, for which more data are available, phylosymbiosis is not universal. Insects, sponges and mammals seem to exhibit phylosymbiosis more consistently than other classes.

Some of the first investigations of phylosymbiosis in the gut were in insects and arachnids. An early study examining congruence between host-associated microbiota and host phylogeny in *Nasonia* wasps found that differences in microbial community composition mirror host evolutionary history and that these results are robust across multiple stages of development[90]. In *Nasonia* species, high host–microorganism specificity at times results in hybrid lethality in hosts[16]. Similarly, within *Cephalotes* ants, differences in gut microbial community composition mirror host phylogeny, and this pattern of phylosymbiosis seems to be maintained by co-diversification[22]. Within termites, major shifts in the composition of the gut microbiome have occurred at major evolutionary transitions — changes in dietary strategy or the gain or loss of eukaryotic symbionts[91]. Thanks in part to this foundational work, there are numerous studies indicating phylosymbiosis in insects. Gut microbiomes of cockroaches, termites, ticks, mosquitoes, ants, wasps and bees all demonstrate patterns of phylosymbiosis[22,91–96]. These symbionts that follow a pattern of phylosymbiosis with their insect hosts include both endosymbionts and 'exosymbionts', and some of these symbiotic relationships are obligate. Some populations of dung beetles also show evidence of phylosymbiosis[97]. However, evidence for phylosymbiosis in the Lepidoptera and Diptera taxa is mixed or absent[92,98–101].

Phylosymbiosis is also common across sponges, in both tropical and cold water coral reef systems[24,102–107]. In a sample of seven closely related sponges, host-associated microbial communities were diverse (47 phyla associated with members of a single genus) and host specific[104]. However, one study suggests that the sponge microbiome reflects the microbiome of the surrounding coral reef community[108], and several taxa of sponges have low microbial abundance[102,103,105,109]. In several studies, the host specificity of the sponge microbiota outweighs geographic variation and variation in environmental conditions, even in low microbial abundance sponges[104,105,110–112].

Phylosymbiosis has been found in a range of mammals[5,23,113,114], including primates[13,75], rodents[11,92,115] and large herbivores[116] (FIG. 1). A study of 60 species of mammals showed a large influence of both the diet and phylogeny on gut microbial community structure; specifically, patterns of gut microbiome similarity mirrored mammalian phylogeny more often than expected by chance, potentially due to co-diversification[5]. Within apes, researchers found congruence between a maximum parsimony tree constructed from gut microbiome composition data and a phylogenetic tree based on host mitochondrial DNA[75]. A reanalysis of the same dataset also found that microbial relationships correlated with host phylogeny in apes, but only when bacterial strains were examined, indicating that recent bacterial evolution is driving this pattern[22].

A key question that results from the phylosymbiosis literature is why evidence for it is so varied. For non-mammalian vertebrate taxa, including birds, amphibians and ray-finned fish, some studies argue for the presence of phylosymbiosis, whereas others report mixed or weak results. For example, whereas two studies found evidence for phylosymbiosis in birds[117,118], a third found no evidence[119]. Importantly, a recent analysis of bats, birds and non-flying animals suggests that phylosymbiosis may be weaker in flying animals than in terrestrial animals[120] (FIG. 1).

Even within animal classes that exhibit consistent patterns of phylosymbiosis, there is clear variation in effect size across studies[6,14,120–122]. In part, these inconsistencies may be a result of variation in methods, such as the use of different metrics of phylosymbiosis or the examination of patterns at distinct host taxonomic scales. Additionally, the microbial taxonomic resolution of studies and assessments of microbial taxonomy rather than function have the potential to influence signals of phylosymbiosis given that closely related and functionally redundant microbial lineages are often assigned as taxonomically distinct entities[123,124]. However, in the existing literature, patterns in the prevalence and strength of phylosymbiosis appear to be only loosely associated with study design. As a result, we must also more directly consider the community assembly processes underlying phylosymbiosis and the extent to which they differ across host taxonomic groups.

Although this type of exploration can and should be applied to specific groups of host taxa within a given class, a more tractable starting point may be examining trends more broadly across the animal kingdom. From this perspective, one important pattern that emerges from the existing phylosymbiosis literature is that phylosymbiosis is less prevalent in host taxa with taxonomically richer microbiomes (FIG. 3). In particular, vertebrates possess richer microbial communities than invertebrates, and phylosymbiosis occurs less frequently in vertebrates. The processes of microbial community assembly described above provide some insight into this pattern. As a microbial community becomes taxonomically (and functionally) richer, it becomes more difficult for processes of microbial community assembly to affect the entire community to the same degree. There are physical limits to the number of microbial lineages that can colonize the germ line, and other forms of vertical transmission are more susceptible to random losses and gains of microbial lineages. Similarly, there are physiological limits to the number of specific signals that immune cells and other host filtering mechanisms can produce to precisely influence colonization and diversification by multiple lineages of microorganisms. As a result, microbial community assembly should become less deterministic and more stochastic as the number

Endosymbionts
Symbiotic, often mutualistic, organisms living inside the tissue of another organism. Symbionts are transmitted either horizontally or vertically, and the symbiosis relationship can be obligatory or not obligatory.

Host specificity
Microbial ability to colonize a host depending on physiological interactions with the host and environmental requirements (for example, pH tolerance, biofilm regulation and polysaccharide utilization loci).

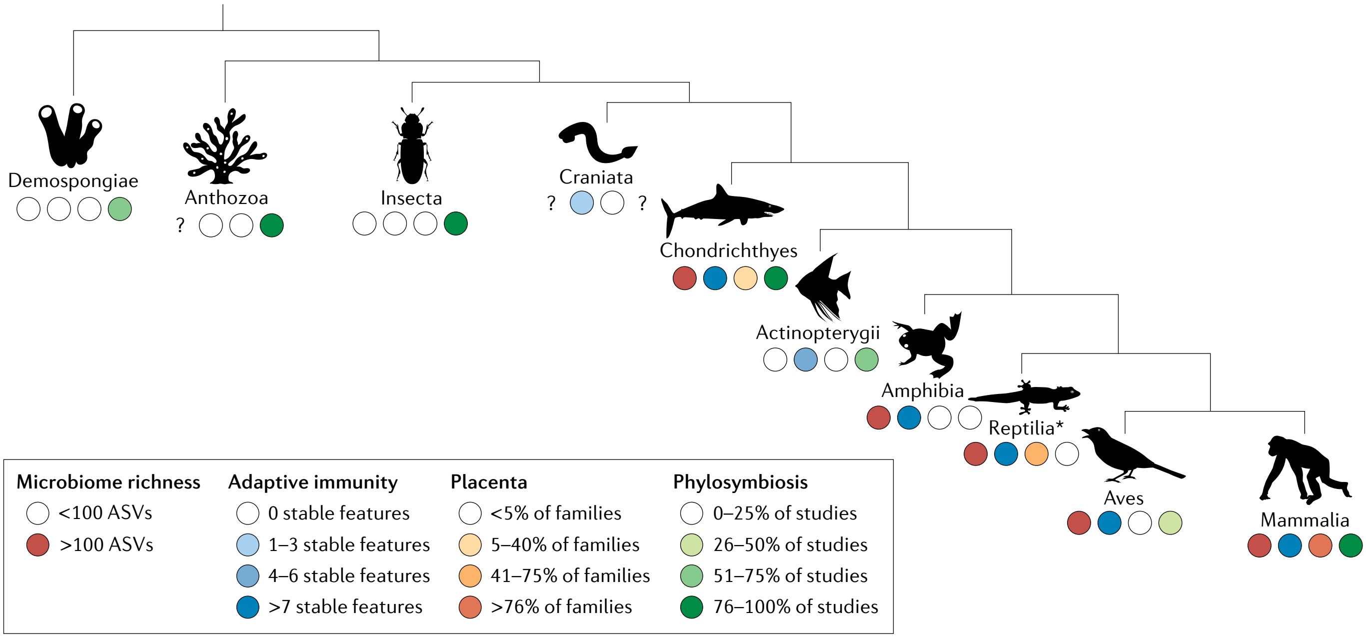


Fig. 3 | **Prevalence of microbial and host immune traits across animal clades.** Mammals are the only class of animals that exhibit a rich microbiome, highly complex adaptive immune systems and prevalent placentation. The adaptive immune traits necessary to tolerate placentas may explain the prevalence of phylosymbiosis in the microbiomes of mammals compared with most other vertebrates. Loss of function of key adaptive immune adaptations (for example, the CNS1 enhancer) in mammals results in loss of pregnancy and loss of gut microbial diversity. This figure illustrates the co-occurrence of microbiome richness, adaptive immune traits, placentas and microbial phylosymbiosis in major evolutionary groups of animals. Only invertebrate classes with at least four studies were included. All vertebrate classes have at least four studies, with the exception of Craniata, Chondrichthyes and Reptilia. Vertebrates generally have richer microbiomes than invertebrates. Among vertebrates, only mammals exhibit a complex adaptive immune system, prevalent placentas and consistent evidence of phylosymbiosis. Microbiome amplicon sequence variant (ASV) richness was calculated from REF.[59] (Supplementary Table 4). Classes missing from that analysis were not assigned a category given the known impact of cross-study methods on microbiome richness. Adaptive immune complexity was categorized by the presence or absence of stable features as described in REF.[140] (Supplementary Table 3). Stable features include thymus, spleen, lymph nodes, somatic hypermutation, class switch recombination, germinal centres, T cell receptors and/or immunoglobulin heavy chain variable regions, major histocompatibility complexes, dendritic cells and follicular dendritic cells. Data from REFS[142,177] were used to determine the approximate percentage of families in a given taxonomic group that exhibit placentas (Supplementary Table 3). For non-avian reptiles, it is important to note that placentas are somewhat prevalent but almost always associated with eggs (asterisk). Evidence supporting phylosymbiosis in a given host taxonomic group was assessed by a literature review (FIG. 1; Supplementary Tables 1,2). A question mark indicates missing data.

of host–microorganism pairwise interactions increases exponentially with microbiome richness.

On the basis of current available data, there are two exceptions to the pattern of reduced phylosymbiosis with increased microbiome richness. First, despite being invertebrates, a subset of sponges have high microbiome richness and exhibit phylosymbiosis[24,102–105] (FIGS 1,3). Because this is not a universal pattern in sponges, it requires more investigation beyond existing knowledge of sponge physiology and host–microorganism interactions. However, it may be linked to host influences on microbial dispersal and selection of microbial lineages as well as microbial influences on host diversification. Specifically, sponges use a mixture of horizontal and vertical transmission, including germ line transmission, to influence microbial dispersal[105]. They also have a robust innate immune system that appears to recognize a higher diversity of non-self molecules[105,106]. Finally, sponge microbiota provide a number of essential metabolic functions for their hosts[24,106,125,126], which has likely led to patterns of host–microorganism co-diversification.

The second exception is that although phylosymbiosis is rare in most vertebrates, it is common in mammals[127,128] (FIGS 1,3). In contrast to sponges, much more is known about mammalian physiology and host–microorganism interactions. As a result, mammals are currently an important clade in which to test the relationship between microbial community assembly processes, microbiome richness and phylosymbiosis. Understanding how deterministic processes of microbial community assembly are maintained in this host clade despite taxonomically and functionally rich microbiomes can provide important insight into the relative importance of various microbial community assembly processes. It may also allow a comparison of the contributions of host versus microbial traits to these processes. Nevertheless, to date, there has been little exploration of the factors that make host–microorganism interactions in mammals different. We outline key areas of interest in the following sections to facilitate more research in this area.

Why do mammals exhibit phylosymbiosis?

There is no a priori reason to believe that microbial strategies influencing microbial community assembly processes differ in mammalian-associated microbial

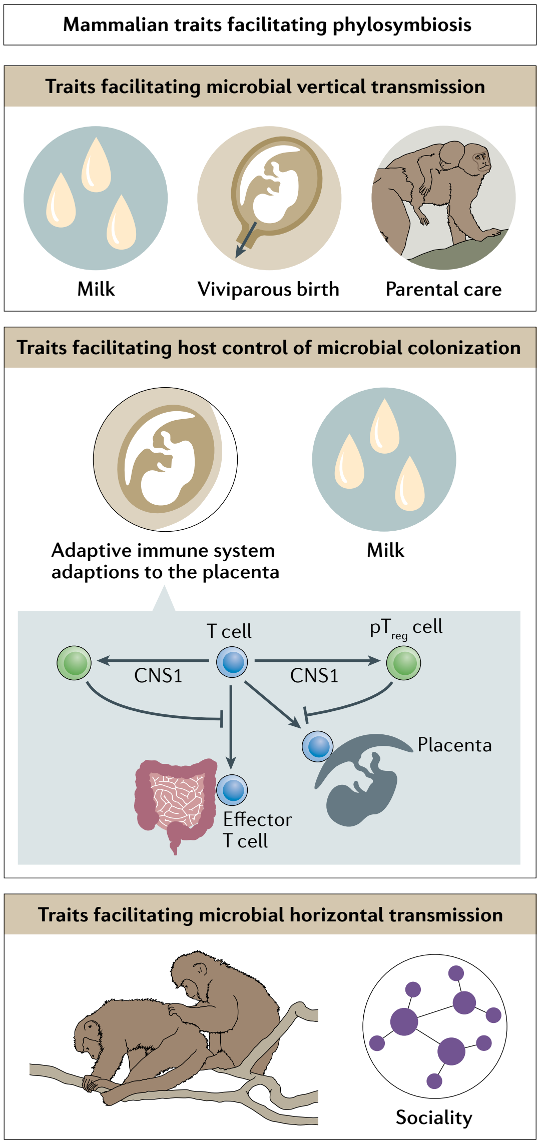


Fig. 4 | **Mammalian traits promoting phylosymbiosis.** Mammals possess a unique suite of traits that facilitates phylosymbiosis by ensuring intergenerational microbial exposure through vertical and horizontal transmission and by strengthening host control of microbial colonization. Mammalian traits facilitating vertical transmission of microorganisms include milk (which is a source of microorganisms and also encourages mother–infant contact), viviparous birth (which facilitates direct contact between the maternal microbiota and the infant body) and parental care (which ensures prolonged physical contact between parents and offspring). Horizontal transmission of microorganisms is facilitated via the social behaviour of many mammals (which results in frequent direct and indirect contact with the microorganisms of conspecifics). Mammalian traits that convey improved host control of microbial colonization include milk (which serves as both a prebiotic and a targeted antimicrobial agent) and adaptive immune system adaptations to the placenta. In the latter case, the CNS1 enhancer, which is unique to mammals, promotes T cell differentiation into peripheral regulatory T (pT$_{reg}$) cells that result in host tolerance of both the placenta and the microbiome.

Monotremes
Egg-laying mammals.

lineages compared with microbial lineages associated with other vertebrate host clades. Instead, it is more likely that evolved mammalian traits result in differences from other vertebrates in host-driven microbial community assembly processes. Mammals are characterized by a suite of traits that are common or universal throughout the class Mammalia and that are absent or rare in other vertebrates. These traits include hair, mammary glands, viviparous birth, parental care of offspring and placentas. Most of these host traits, with perhaps the exception of hair, strongly reinforce at least two of the processes driving phylosymbiosis (FIG. 4): microbial dispersal and environmental selection of microbial lineages. Although other vertebrates may exhibit some of these host traits, only mammals exhibit them all. Therefore, the cumulative effects of these host traits may offset the stochasticity associated with the dispersal and selection of rich microbiomes such that community assembly is more deterministic and less stochastic in mammals compared with other vertebrates. In short, the unique suite of mammalian traits may be key to understanding the strong signal of phylosymbiosis in mammals.

Mammalian traits increasing the likelihood of deterministic microbial dispersal. The potential for vertical transmission of microorganisms is greatly increased by viviparity in mammals (FIG. 4). Although this trait is not unique to mammals, it is almost universal (with the exception of monotremes), and examples of it outside the class are limited. Most other vertebrates are oviparous. As a result of viviparity, one of the first major microbial exposures that mammalian infants come into contact with is the maternal vaginal and faecal microbiota[36,37,84]. Because the infant body comes into direct contact with these microorganisms, the potential for rapid, faithful transmission is high. By contrast, although the eggs of oviparous vertebrates come into contact with maternal and/or paternal microorganisms, there is also substantial environmental contact. Therefore, there is increased potential for stochastic microbial colonization of the egg, and the egg microbiota is unlikely to be dominated by parental microorganisms[34]. Furthermore, in some cases, parental behaviours constrain later contact between parents and their eggs and/or infants, which can further reduce transmission of maternal and/or paternal microorganisms. For example, whereas birds incubate eggs, turtles provide no parental care after depositing eggs, reducing the potential for vertical transmission[129]. External fertilization, as observed in marine and aquatic animals such as fish, further reduces the potential for vertical transmission, except in the case of germ line inoculation, which has been reported in some fish, sponges and bivalves[28–33].

In addition to increasing the deterministic nature of microbial dispersal in mammals, the vertical transmission promoted by viviparity likely also triggers priority effects in the developing mammalian microbiota. By ensuring early exposure of infants to specific parental microbial communities, viviparity may make it less likely that other microbial taxa can stochastically colonize the infant subsequently. For example, programming of the host immune system by certain microbial taxa can lead to the exclusion of other microbial taxa[52]. Nevertheless, empirical data describing the extent to which viviparity results in priority effects in the gut microbiota and identifying the mechanisms through which these priority effects operate are lacking.

Mammalian parental behaviours also increase opportunities for vertical transmission of microorganisms (FIG. 4). As with viviparity, parental care is not unique to mammals. Birds, for example, engage in extensive parental care. However, mammalian parental care differs from that of other vertebrates, particularly with respect to lactation. All female mammals have mammary glands, which produce milk to feed offspring. Milk is another source of microorganisms that is delivered directly from the maternal body to the infant gut, allowing efficient vertical transmission[130,131]. Similar dynamics could exist in other vertebrates. For example, bird predigestion of food for offspring or production of crop milk[132] could result in microbial transmission. However, this behaviour is also somewhat unique among vertebrates, and little is known about the microbial composition of foods passed from parents to offspring in birds. Furthermore, nursing involves direct, prolonged physical contact between mammalian mothers and offspring, which further promotes vertical transmission of microorganisms from maternal skin and other body sites.

Beyond lactation, parental proximity sets mammalian parental care apart from that of other vertebrates. During infancy, mammals commonly carry their young and/or remain in close physical contact, which promotes microbial vertical transmission. Even after infant mammals have been weaned, many continue to exist in close physical contact with their mothers into adulthood. For example, female juvenile primates in matrilineal species, such as macaques (*Macaca* spp.), interact closely with their mothers well after they have been weaned. As a result, the window of time during which parental care can facilitate vertical transmission is extended.

As described for viviparity, milk and parental care can also trigger priority effects in the developing infant microbiota as a result of the vertical transmission processes they support. The probiotic properties of milk ensure that specific microbial lineages are introduced into the gut almost immediately after birth. Likewise, the physical contact involved with mammalian parental care may increase the probability that the first microorganisms with which infants are colonized are parental. This may reduce the stochasticity of subsequent microbial colonization as a result of intermicrobial competition or programming of the host immune system. However, as with viviparity, more data are needed to better understand the relative importance of these processes.

Although not a unique or a universal trait, many mammals live in complex social groups, which promotes horizontal transmission of microorganisms (FIG. 4). Depending on the host species, therefore, microorganisms have the potential to be transferred between a few siblings or among large groups of kin and non-kin conspecifics. Microorganisms have been shown to be transmitted via these pathways in some mammalian taxa[44,45], although the effects on the developing infant gut microbiome are understudied. However, it seems likely that infants living in large groups of conspecifics have a higher probability of being exposed to species-specific microorganisms than infants that are not[133,134]. For example, human infants with larger social networks have a higher diversity of skin microorganisms[134]. The same dynamics are also likely to promote maintenance of host species-specific microbial associations across lifespans. Conspecific primates have more similar gut microbiomes when they spend more time in close proximity or physical contact[44,45]. Whether this pattern is stronger in mammalian host species with larger or more cohesive social groups remains to be seen. In insects, however, sociality seems to be over-represented in the species that exhibit phylosymbiosis[135].

Mammalian traits increasing control over microbial selection. In addition to facilitating microbial dispersal via vertical transmission, milk imposes selective pressure on the developing mammalian microbiome (FIG. 4). Milk is a source of prebiotics that serve as substrates for specific bacteria[136,137]. It also contains a range of immune factors and antimicrobials that can directly influence the survival of different microbial lineages. For example, IgA is secreted in breast milk and will bind to many non-maternal microbial lineages in the infant gut[138]. This property allows milk to act as a filter for microorganisms establishing themselves in the infant gut by creating environmental conditions that favour a subset of microorganisms. Additionally, milk provides maternal immune protection for infant mammals across a much longer period than is observed in other animals. Invertebrates receive essentially no maternal immune protection, and birds and reptiles receive it only within their eggs[139]. As a result, infant mammals may have stronger immune influences on microbial selection during early life.

Eutherian placenta–immune system interactions are also likely to have a role in host selection of the microbiota (FIG. 4). In general, the mammalian immune system is considered to be more complex than that of many other vertebrate classes[140], which may contribute to stronger host filtering of microorganisms. In particular, T_{reg} cells and T helper 17 cells have an active role in shaping the gut microbiome by suppressing immune responses to gut microorganisms[141]. However, other vertebrates, such as birds, have complex immune systems with similar T cell functionality[140] and do not exhibit strong patterns of phylosymbiosis[119,120]. Here, examining the evolution of the mammalian placenta and adaptive immune system together provides additional insight (FIG. 3). Although some other animals have placentas, their structure is different from that of mammals in that they do not invade maternal tissue as extensively. Because of its invasive structure, the evolution of the mammalian placenta required key changes to the host adaptive immune system that increased tolerance to non-self cells (BOX 1). As a result, peripheral T_{reg} (pT_{reg}) cells function uniquely in mammals compared with other vertebrates[142,143]. These changes not only confer tolerance of invasive placentas but they also appear to have paved the way for complex mammalian microbiota. pT_{reg} cells have receptors for microbial metabolites and can block dendritic cells

Matrilineal species
Species in which females remain in their natal group, males disperse, dominance rank is inherited by females from their mothers and social bonds between closely related females are strong.

Prebiotics
Dietary items which provide substrates for bacterial growth in the digestive tract.

Box 1 | Evolution of the placenta

The evolution of the mammalian placenta is likely to have been a key development in vertebrate host–microorganism interactions. Placentas are not unique to mammals, but the combination of an invasive placenta with a complex adaptive immune system is[142]. Therefore, the mammalian immune system has evolved unique functions that confer tolerance to the placenta. These immune adaptations also appear to be critical in governing host–microorganism interactions in mammals.

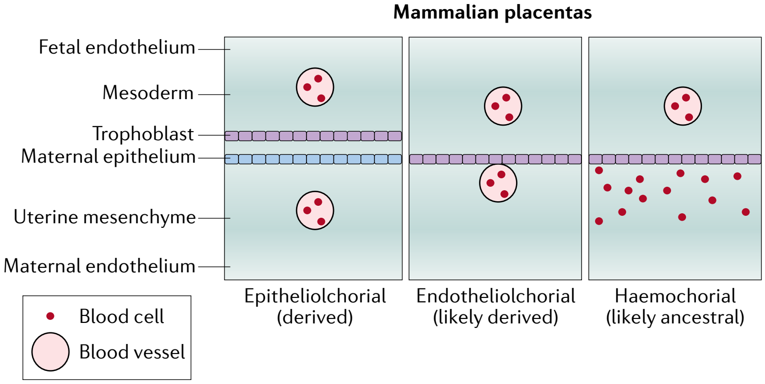
Every eutherian mammal possesses one of three main types of placentas: epitheliochorial, endotheliochorial and haemochorial (see the figure). Epitheliochorial placentas have two layers of maternal cells separating maternal blood and fetal cells and are therefore minimally invasive. Endotheliochorial placentas are moderately invasive and have one layer of maternal cells separating maternal blood from fetal cells. Finally, haemochorial placentas bathe fetal cells in maternal blood directly[178] and are considered the most invasive type of placenta. Highly invasive haemochorial placentas are believed to be the ancestral state in mammals[179,180], although moderately invasive endotheliochorial placentas could also be ancestral depending on where the mammalian phylogeny is rooted.

To tolerate an invasive placenta, the mammalian adaptive immune system had to evolve a stronger regulatory system. In particular, the adaptive immune system needed to be able to reduce the activity of T helper cells[142]. As a result, the mammalian adaptive immune system has a uniquely expanded population of regulatory T cells that can inhibit the activity of T helper cells. Specifically, there is an expansion of peripheral regulatory T (pT_{reg}) cells (outside the thymus) that can block the activity of dendritic cells that sense non-self signalling molecules[143,145]. This activity protects the placenta and the fetus from maternal immune activation. When pT_{reg} cells are depleted, pregnancy termination occurs[143].

Although, genes for cells analogous to mammalian pT_{reg} cells are present in other vertebrates, they are inactive. A unique variant of the CNS1 enhancer has allowed their expanded activity in mammals[143,178]. Specifically, it enhances the *FOXP3* gene, which stimulates differentiation of pT_{reg} cells. Therefore, although the immune systems of some other vertebrates could exhibit tolerance functions similar to those of mammals, they do not. The evolution of an invasive placenta appears to have triggered this adaptation.

The tolerogenic function of pT_{reg} cells also facilitate the maintenance of a rich gut microbial community. When pT_{reg} cells are experimentally depleted in mice, depletion of the gut microbiome occurs[144]. Similarly, vertebrates lacking pT_{reg} cell function do not consistently exhibit host species specificity of the microbiome.

In conclusion, the evolution of an invasive placenta is a key event in the development of host–microorganism interactions in mammals. To tolerate the placenta, the mammalian immune system had to evolve increased regulatory control via production of pT_{reg} cells. These immune changes appear to have simultaneously allowed improved mammalian tolerance and control of the microbiota.



Sympatric terrestrial species
Distinct species living in the same terrestrial habitat.

from responding to these metabolites[144,145]. This interaction stops host responses that would exclude specific microorganisms from the host body. As a result, when pT_{reg} cell function is knocked out in mice, approximately 80% of the gut mucosal microbiome is lost[144]. Therefore, the pT_{reg} cell function that evolved to protect the placenta and fetus in mammals can also protect certain members of the microbiota. Additionally, mammalian B cells and their products are more diverse and specific than those of other vertebrates[57,58]. B cells can influence the microbiota directly via antibody production, but they also regulate T cell differentiation and function[146,147]. These unique B cell and T cell traits in mammals likely allow the mammalian immune system to exercise stronger selection on the microbiota than other vertebrate immune systems. Bidirectional interactions between gut microorganisms and the mammalian adaptive immune system may also result in increased host specificity or fidelity of microorganisms and/or co-diversification of hosts and certain microbial taxa. However, more comparative data are necessary to robustly explore this hypothesis.

Relative importance of drivers in mammals

If phylosymbiosis in mammals is an emergent property resulting from a combination of unique traits that together promote more deterministic patterns of microbial dispersal and selection, we should observe a weakening of phylosymbiosis when one of these traits is altered. Additionally, assessing the extent to which alterations in a given trait affect the strength of phylosymbiosis will provide insight into the relative importance of that trait in maintaining phylosymbiosis. Although these dynamics have not been explicitly tested in the literature, enough data exist to begin to examine general patterns.

Breakdown of microbial dispersal. Variation in processes of microbial dispersal is relatively common throughout the mammalian literature. The most well-studied examples come from humans and captive animals, in which interruption of transmission pathways has been shown to alter early-life microbial exposures. For example, infant mammals born via caesarean delivery, fed with formula and/or isolated from social contact exhibit distinct microbiomes compared with those that are not[148,149]. Because early-life interventions such as caesarean delivery, formula feeding and social isolation do not occur in wild animals, it is more difficult to explore the influence of microbial dispersal processes on phylosymbiosis in this context. However, data from monotremes could provide some insight. Additionally, variation in the duration and type of parental care could influence the strength of transmission pathways across host taxa. For example, we might expect host taxa with early weaning ages to exhibit weaker patterns of phylosymbiosis. Similarly, mammals that live in smaller social groups and thus have reduced contact with conspecifics might experience reduced horizontal microbial transmission[78,79]. Finally, to some extent, host environments can alter microbial dispersal patterns in ways that increase stochasticity and weaken phylosymbiosis. For instance, when closely related host species share an environment, similar exposure to environmental microorganisms as well as microbial transmission between host species might reduce signals of phylosymbiosis. Within lemurs, sympatric terrestrial species have more similar gut microbiomes than sympatric arboreal species, likely due to increased horizontal transmission and similarity in environmental contact[150]. Similarly, in a hybrid zone of yellow and Anubis baboons, soil properties strongly predict the composition of the gut microbiome but host genetics do not[151].

The effects of variation in mammalian traits influencing microbial dispersal on phylosymbiosis have yet to be tested explicitly in any context. However, data indicating an eventual recovery of the microbiome of human infants with altered early-life microbial exposures suggest effects on long-term microbial community assembly are limited[152,153]. If this is true, the redundancy of mammalian traits that facilitate microbial community assembly is likely to be an important explanatory factor. For example, human infants born by caesarean delivery may still receive breast milk, or infants who are formula fed likely still receive substantial parental care. These alternative pathways for microbial dispersal may maintain the deterministic nature of microbial community assembly despite interruptions to other dispersal pathways. Furthermore, host selection mechanisms continue to operate even when dispersal is compromised. Therefore, it is also possible that host selection has a stronger role in microbial community assembly than microbial dispersal. Nevertheless, because selection can act only on microbial lineages that have dispersed, the opposite may also be true. More research is necessary to distinguish between these alternatives.

Altering host control of microbial selection. To some extent, it is more difficult to find examples of altered host selection than altered microbial dispersal in mammals. However, the literature on immunity and the microbiota provides an important foundation. Suppression of the adaptive immune system may decrease the strength of host selectivity, increasing the diversity of microorganisms that can colonize the host. For example, deficiencies in specific immune genes lead to increases in the abundance of certain bacterial taxa[154,155]. Mice without mature lymphocytes have increased rates of colonization with *Akkermansia muciniphila* compared with wild-type mice[60]. Finally, robust host filtering of the gut microbiota occurs in zebrafish only with normal adaptive immune function[156]. Although immune alterations of this magnitude are likely to be rare in wild animals, variation in immune function between orders or genera of mammals could contribute to differences in host control of microbial colonization and phylosymbiosis. It is also known that populations of host species can experience reductions in immune function as a result of energetic trade-offs during periods of energetic constraint, such as reduced food availability[157,158]. Therefore, the strength of phylosymbiosis could vary for groups of host taxa across habitats or seasons.

Although these hypotheses have not been tested directly, data from bats provide some preliminary support. Despite belonging to the class Mammalia, bats do not exhibit strong phylosymbiosis and have high interindividual microbiome variation within a species[120,122,159,160]. Like other mammals, bats have viviparous birth, lactate and often live in large social groups. Therefore, there is no reason to believe that microbial dispersal differs from that of other mammals. However, bats have several physiological adaptations that result in increased immune tolerance of microorganisms. To minimize the physiological damage associated with high metabolic rates during flight, bats have reduced oxidative stress responses, reduced DNA damage detection ability and increased DNA damage repair responses[161]. These adaptations generally reduce immune reactivity to pathogens, particularly viral pathogens, but they may also alter host filtering of microorganisms more generally. Additionally, weakened tight junctions between enterocytes increase paracellular nutrient absorption to support the energetics of flight but also allow more microbial transfer across the gut epithelium[162–165]. To avoid constant immune activation as a result of these exposures, bats have increased immune tolerance and resilience to microbial infections[166–168]. Specifically, bats have lost the PYHIN gene family[168,169] involved in detecting microbial DNA[170,171], have attenuated interferon and cytokine responses[168,172,173] and do not show a normal innate immune response to a lipopolysaccharide challenge[174]. These traits make bats ideal reservoirs of viral disease[166], but they likely also alter host filtering of microorganisms in ways that weaken phylosymbiosis. Recent changes in host ecology can also weaken phylosymbiosis as a result of host physiological adaptations that do not follow host phylogeny. For example, dietary convergence can lead to gut microbiota convergence, irrespective of host phylogeny[6]. Ant-eating mammals from disparate families show convergent evolution in their microbiota that is likely driven by dietary and physiological similarity[7]. Similarly, individuals from creosote-feeding populations of two different species of wood rats have more similar gut microbiomes than creosote-feeding and creosote-naive populations of the same species when challenged with a creosote-containing diet[175]. Larger-scale changes in host ecology can also alter patterns of phylosymbiosis. The gut microbiomes of primates follow a pattern of phylosymbiosis more broadly[13], but the human gut microbiota clusters with that of baboons more than that of more closely related apes because our recent hominin ancestors evolved in an ecological context that is more baboon-like than ape-like[14,176]. In these cases, microbiomes are still host species specific and vary with host physiology. However, host physiology does not mirror host phylogeny. Finally, bats do not exhibit strong phylosymbiosis and also possess flight-associated adaptations in digestive anatomy that are likely to alter microbial filtering. Their reduced gut sizes and quick gut passage rates minimize weight while simultaneously allowing frequent feeding to meet the high energy needs of flight[164]. These short gut retention times may cause rapid gut microbial turnover, increasing the importance of environmental acquisition and transient microbial colonization. Indeed, less transient members of the gut microbial community, such as mucosal-associated bacteria, show a stronger signal of phylogeny in bats[159].

Together, these examples suggest that variation in host traits that affect selection of microbial lineages may have bigger impacts on patterns of microbial community assembly and phylosymbiosis than variation in host traits that affect microbial dispersal. Nevertheless, more research is necessary to robustly test these patterns and more specifically identify the host–microorganism

Sympatric arboreal species
Distinct species living in the same arboreal habitat.

Host selectivity
Host ability to limit the microbial strains that associate with it via various physiological filters (for example, immune system, gut anatomy and diet/nutrients).

interactions of interest. Given that mammalian phylosymbiosis appears more broadly among vertebrates, it remains likely that the cumulative effects of multiple mammalian traits on multiple microbial community assembly processes are necessary to support it. Studies in both wild and laboratory settings that target individual mammalian traits contributing to microbial community assembly will be crucial for advancing knowledge in this area.

Conclusions

Phylosymbiosis provides an effective foundation for elucidating generalized rules for host–microorganism interactions. Despite clear gaps, current data testing for phylosymbiosis across the animal kingdom suggest that it may become less prevalent as microbiomes become taxonomically richer. This pattern provides crucial insight into the scales at which different microbial community assembly processes operate. Mammals, which appear to be an exception among microbially rich vertebrates, provide additional leverage for assessing the relative importance of distinct community assembly processes. We hypothesize that the prevalence of phylosymbiosis in mammals is a result of a unique combination of mammalian traits, including viviparity, milk, parental care, sociality and immune adaptations to the placenta. Although some of these traits can be identified in other vertebrates, in mammals they compound to strengthen host control of microbial dispersal and selection. When one or more of these traits are altered, phylosymbiosis can break down, and evidence from bats suggests that immune adaptations that alter host selection for microorganisms are particularly important. More data from host taxonomic groups such as marsupials and monotremes, as well as host taxa with variation in the type and length of parental care and with variation in immune function, will allow further testing of this hypothesis.

More generally, although the concept of phylosymbiosis is ubiquitous in the microbiome literature, substantial work remains before we can develop generalizable principles of host-associated microbial community assembly. Numerous animal classes lack basic information about the presence or absence of phylosymbiosis, and the relative importance of the community assembly processes, including dispersal, selection, co-diversification and ecological drift, may differ across host taxa. Furthermore, current approaches are often limited to taxonomic descriptions of microbiome composition that may obscure patterns of phylosymbiosis by either separating phylogenetically and functionally redundant microbial lineages or clustering them at high taxonomic levels that lead to overly conserved patterns of association with hosts[23,123]. Moving forward, analyses should incorporate approaches such as ecophylogenetics[123] to reduce this weakness and should simultaneously increase microbial taxonomic resolution using alternative genetic markers and/or shotgun metagenomic data and associated microbial genomes. Finally, phylosymbiosis research should move beyond binary questions of diet versus phylogeny and instead evaluate how strong the signal of phylogeny is, in what contexts and why. Are both controlled microbial dispersal and host control of microbial selection necessary, or is one of those processes sufficient provided it is robust enough? How often do these processes break down in nature, and do these breakdowns interrupt phylosymbiosis? If so, what is the impact on host health, ecology and even evolution? Addressing these questions will allow us to continue to advance our understanding of host–microorganism interactions across the animal kingdom.

Published online 27 May 2021

1. Al Nabhani, Z. & Eberl, G. Imprinting of the immune system by the microbiota early in life. *Mucosal Immunol.* **13**, 183–189 (2020).
2. Pronovost, G. N. & Hsiao, E. Y. Perinatal interactions between the microbiome, immunity, and neurodevelopment. *Immunity* **50**, 18–36 (2019).
3. Sylvia, K. E. & Demas, G. E. A gut feeling: microbiome-brain-immune interactions modulate social and affective behaviors. *Horm. Behav.* **99**, 41–49 (2018).
4. Visconti, A. et al. Interplay between the human gut microbiome and host metabolism. *Nat. Commun.* **10**, 4505 (2019).
5. Ley, R. E. et al. Evolution of mammals and their gut microbes. *Science* **320**, 1647–1651 (2008). **This work examines the gut microbiomes of 60 species of mammals, finding that both host diet and phylogeny strongly influenced gut microbial composition**.
6. Muegge, B. D. et al. Diet drives convergence in gut microbiome functions across mammalian phylogeny and within humans. *Science* **332**, 970–974 (2011).
7. Delsuc, F. et al. Convergence of gut microbiomes in myrmecophagous mammals. *Mol. Ecol.* **23**, 1301–1317 (2014).
8. Song, S. J. et al. Is there convergence of gut microbes in blood-feeding vertebrates? *Phil. Trans. R. Soc. B* **374**, 20180249 (2019).
9. McKenney, E. A., Maslanka, M., Rodrigo, A. & Yoder, A. D. Bamboo specialists from two mammalian orders (Primates, Carnivora) share a high number of low-abundance gut microbes. *Microb. Ecol.* **76**, 272–284 (2018).
10. Amato, K. R. et al. The gut microbiota appears to compensate for seasonal diet variation in the wild black howler monkey (*Alouatta pigra*). *Microb. Ecol.* **69**, 434–443 (2015).
11. Kohl, K. D., Varner, J., Wilkening, J. L. & Dearing, M. D. Gut microbial communities of American pikas (*Ochotona princeps*): evidence for phylosymbiosis and adaptations to novel diets. *J. Anim. Ecol.* **87**, 323–330 (2018).
12. Moeller, A. H. & Sanders, J. G. Roles of the gut microbiota in the adaptive evolution of mammalian species. *Phil. Trans. R. Soc. B* **375**, 20190597 (2020).
13. Amato, K. R. et al. Evolutionary trends in host physiology outweigh dietary niche in structuring primate gut microbiomes. *ISME J.* **13**, 576–587 (2019).
14. Amato, K. R. et al. Convergence of human and old world monkey gut microbiomes demonstrates the importance of human ecology over phylogeny. *Genome Biol.* **20**, 201 (2019).
15. Mazel, F. et al. Is host filtering the main driver of phylosymbiosis across the tree of life? *mSystems* **3**, e00097-18 (2018).
16. Brucker, R. M. & Bordenstein, S. R. The hologenomic basis of speciation: gut bacteria cause hybrid lethality in the genus *Nasonia*. *Science* **341**, 667–669 (2013). **The study shows that relationships between the microbial communities of *Nasonia* wasps recapitulate host phylogeny when different species are reared in the same environmental conditions**.
17. Bordenstein, S. R. & Theis, K. R. Host biology in light of the microbiome: ten principles of holobionts and hologenomes. *PLoS Biol.* **13**, e1002226 (2015).
18. Koskella, B. & Bergelson, J. The study of host–microbiome (co)evolution across levels of selection. *Phil. Trans. R. Soc. B* **375**, 20190604 (2020).
19. Lim, S. J. & Bordenstein, S. R. An introduction to phylosymbiosis. *Proc. R. Soc. B* **287**, 20192900 (2020).
20. O'Brien, P. A. et al. Diverse coral reef invertebrates exhibit patterns of phylosymbiosis. *ISME J.* **14**, 2211–2222 (2020).
21. Kohl, K. D. Ecological and evolutionary mechanisms underlying patterns of phylosymbiosis in host-associated microbial communities. *Phil. Trans. R. Soc. B* **375**, 20190251 (2020).
22. Sanders, J. G. et al. Stability and phylogenetic correlation in gut microbiota: lessons from ants and apes. *Mol. Ecol.* **23**, 1268–1283 (2014).
23. Groussin, M. et al. Unraveling the processes shaping mammalian gut microbiomes over evolutionary time. *Nat. Commun.* **8**, 14319 (2017).
24. Thomas, T. et al. Diversity, structure and convergent evolution of the global sponge microbiome. *Nat. Commun.* **7**, 11870 (2016).
25. Apprill, A. et al. Marine mammal skin microbiotas are influenced by host phylogeny. *R. Soc. Open Sci.* **7**, 192046 (2020).
26. Bird, A. K., Prado-Irwin, S. R., Vredenburg, V. T. & Zink, A. G. Skin microbiomes of California terrestrial salamanders are influenced by habitat more than host phylogeny. *Front. Microbiol.* **9**, 442 (2018).
27. Doane, M. P. et al. The skin microbiome of elasmobranchs follows phylosymbiosis, but in teleost fishes, the microbiomes converge. *Microbiome* **8**, 93 (2020).
28. Russell, S. L., Chappell, L. & Sullivan, W. in *Current Topics in Developmental Biology* vol. 135 (ed. Lehmann, R.) 315–351 (Academic, 2019).
29. Russell, S. L., McCartney, E. & Cavanaugh, C. M. Transmission strategies in a chemosynthetic symbiosis: detection and quantification of symbionts in host tissues and their environment. *Proc. R. Soc. B* **285**, 20182157 (2018).

30. Usher, K. M., Kuo, J., Fromont, J. & Sutton, D. C. Vertical transmission of cyanobacterial symbionts in the marine sponge *Chondrilla australiensis* (Demospongiae). *Hydrobiologia* **461**, 9–13 (2001).
31. Usher, K. M., Sutton, D. C., Toze, S., Kuo, J. & Fromont, J. Inter-generational transmission of microbial symbionts in the marine sponge *Chondrilla australiensis* (Demospongiae). *Mar. Freshw. Res.* **56**, 125–131 (2005).
32. Nyholm, S. V. In the beginning: egg–microbe interactions and consequences for animal hosts. *Phil. Trans. R. Soc. B* **375**, 20190593 (2020).
33. Funkhouser, L. J. & Bordenstein, S. R. Mom knows best: the universality of maternal microbial transmission. *PLoS Biol.* **11**, e1001631 (2013).
34. Perlmutter, J. I. & Bordenstein, S. R. Microorganisms in the reproductive tissues of arthropods. *Nat. Rev. Microbiol.* **18**, 97–111 (2020).
35. Hosokawa, T., Kikuchi, Y., Shimada, M. & Fukatsu, T. Symbiont acquisition alters behaviour of stinkbug nymphs. *Biol. Lett.* **4**, 45–48 (2008).
36. Dominguez-Bello, M. G. et al. Delivery mode shapes the acquisition and structure of the initial microbiota across multiple body habitats in newborns. *Proc. Natl Acad. Sci USA* **107**, 11971–11975 (2010).
37. Mitchell, C. et al. Delivery mode impacts newborn gut colonization efficiency. Preprint at *bioRxiv* https://doi.org/10.1101/2020.01.29.919993 (2020).
38. Ferretti, P. et al. Mother-to-infant microbial transmission from different body sites shapes the developing infant gut microbiome. *Cell Host Microbe* **24**, 133–145.e5 (2018).
This study uses longitudinal sampling of 25 mother–infant pairs to demonstrate persistent colonization of the infant by maternal gut microbial strains and emphasizes the importance of mother–infant microbial transmission.
39. Brandl, H. B. et al. Composition of bacterial assemblages in different components of reed warbler nests and a possible role of egg incubation in pathogen regulation. *PLoS ONE* **9**, e114861 (2014).
40. Kyle, G. Z. & Kyle, P. D. *Rehabilitation and Conservation of Chimney Swifts* (Driftwood Wildlife Association, 2004).
41. Wang, Y. & Rozen, D. E. Gut microbiota colonization and transmission in the burying beetle *Nicrophorus vespilloides* throughout development. *Appl. Environ. Microbiol.* **83**, e03250-16 (2017).
42. Hosokawa, T. et al. Diverse strategies for vertical symbiont transmission among subsocial stinkbugs. *PLoS ONE* **8**, e65081 (2013).
43. Parker, E. S., Dury, G. J. & Moczek, A. P. Transgenerational developmental effects of species-specific, maternally transmitted microbiota in *Onthophagus* dung beetles: host-symbiont interactions in dung beetles. *Ecol. Entomol.* **44**, 274–282 (2019).
44. Perofsky, A. C., Lewis, R. J., Abondano, L. A., Di Fiore, A. & Meyers, L. A. Hierarchical social networks shape gut microbial composition in wild Verreaux's sifaka. *Proc. R. Soc. B* **284**, 20172274 (2017).
45. Tung, J. et al. Social networks predict gut microbiome composition in wild baboons. *eLife* **4**, e05224 (2015).
This article shows that both shared social group membership and the frequency of social interaction predicted how similar the gut microbiome composition and function of two individuals are.
46. Song, S. J. et al. Cohabiting family members share microbiota with one another and with their dogs. *eLife* **2**, e00458 (2013).
47. Moeller, A. H. et al. Dispersal limitation promotes the diversification of the mammalian gut microbiota. *Proc. Natl Acad. Sci. USA* **114**, 201700122 (2017).
48. Chandler, J. A., Lang, J. M., Bhatnagar, S., Eisen, J. A. & Kopp, A. Bacterial communities of diverse *Drosophila* species: ecological context of a host–microbe model system. *PLoS Genet.* **7**, e1002272 (2011).
49. Franzenburg, S. et al. Distinct antimicrobial peptide expression determines host species-specific bacterial associations. *Proc. Natl Acad. Sci. USA* **110**, e3730–e3738 (2013).
50. McLoughlin, K., Schluter, J., Rakoff-Nahoum, S., Smith, A. L. & Foster, K. R. Host selection of microbiota via differential adhesion. *Cell Host Microbe* **19**, 550–559 (2016).
51. Schluter, J. & Foster, K. R. The evolution of mutualism in gut microbiota via host epithelial selection. *PLoS Biol.* **10**, e1001424 (2012).
52. Chu, H. & Mazmanian, S. K. Innate immune recognition of the microbiota promotes host-microbial symbiosis. *Nat. Immunol.* **14**, 668–675 (2013).
53. Brugman, S. et al. T lymphocytes control microbial composition by regulating the abundance of Vibrio in the zebrafish gut. *Gut Microbes* **5**, 737–747 (2014).
54. Dimitriu, P. A. et al. Temporal stability of the mouse gut microbiota in relation to innate and adaptive immunity. *Environ. Microbiol. Rep.* **5**, 200–210 (2013).
55. Kawamoto, S. et al. Foxp3^{+} T cells regulate immunoglobulin A selection and facilitate diversification of bacterial species responsible for immune homeostasis. *Immunity* **41**, 152–165 (2014).
56. McFall-Ngai, M. J. Care for the community. *Nature* **445**, 153–153 (2007).
This article argues that the adaptive immune system may have evolved in part to regulate interactions between vertebrate hosts and commensal microorganisms given the higher-diversity microbiomes observed in vertebrates.
57. Hsu, E. Mutation, selection, and memory in B lymphocytes of exothermic vertebrates. *Immunol. Rev.* **162**, 25–36 (1998).
58. Parra, D., Takizawa, F. & Sunyer, J. O. Evolution of B cell immunity. *Ann. Rev. Anim. Biosci.* **1**, 65–97 (2013).
59. Woodhams, D. C. et al. Host-associated microbiomes are predicted by immune system complexity and climate. *Genome Biol.* **21**, 23 (2020).
60. Zhang, H., Sparks, J. B., Karyala, S. V., Settlage, R. & Luo, X. M. Host adaptive immunity alters gut microbiota. *ISME J.* **9**, 770–781 (2015).
This article demonstrates that genetically immunodeficient mice have a distinct microbiome and distinct microbial developmental trajectory compared with wild-type mice.
61. Logan, S. L. et al. The *Vibrio cholerae* type VI secretion system can modulate host intestinal mechanics to displace gut bacterial symbionts. *Proc. Natl Acad. Sci. USA* **115**, E3779–E3787 (2018).
62. Macfarlane, G. T., Macfarlane, S. & Gibson, G. R. Validation of a three-stage compound continuous culture system for investigating the effect of retention time on the ecology and metabolism of bacteria in the human colon. *Microb. Ecol.* **35**, 180–187 (1998).
63. Schlomann, B. H., Wiles, T. J., Wall, E. S., Guillemin, K. & Parthasarathy, R. Bacterial cohesion predicts spatial distribution in the larval zebrafish intestine. *Biophys. J.* **115**, 2271–2277 (2018).
64. Godoy-Vitorino, F. et al. Comparative analyses of foregut and hindgut bacterial communities in hoatzins and cows. *ISME J.* **6**, 531–541 (2012).
65. Poole, A. C. et al. Human salivary amylase gene copy number impacts oral and gut microbiomes. *Cell Host Microbe* **25**, 553–564.e7 (2019).
66. Kohl, K. D., Sadowska, E. T., Rudolf, A. M., Dearing, M. D. & Koteja, P. Experimental evolution on a wild mammal species results in modifications of gut microbial communities. *Front. Microbiol.* **7**, 634 (2016).
67. Barroso-Batista, J., Demengeot, J. & Gordo, I. Adaptive immunity increases the pace and predictability of evolutionary change in commensal gut bacteria. *Nat. Commun.* **6**, 8945 (2015).
This article shows that evolution of *Escherichia coli* is slower in genetically immunocompromised mice than in wild-type mice.
68. Foster, K. R., Schluter, J., Coyte, K. Z. & Rakoff-Nahoum, S. The evolution of the host microbiome as an ecosystem on a leash. *Nature* **548**, 43–51 (2017).
69. Martens, E. C., Chiang, H. C. & Gordon, J. I. Mucosal glycan foraging enhances fitness and transmission of a saccharolytic human gut bacterial symbiont. *Cell Host Microbe* **4**, 447–457 (2008).
70. Zhao, S. et al. Adaptive evolution within gut microbiomes of healthy people. *Cell Host Microbe* **25**, 656–667.e8 (2019).
71. Garud, N. R. & Pollard, K. S. Population genetics in the human microbiome. *Trends Genet.* **36**, 53–67 (2020).
72. Kwong, W. K., Engel, P., Koch, H. & Moran, N. A. Genomics and host specialization of honey bee and bumble bee gut symbionts. *Proc. Natl Acad. Sci. USA* **111**, 11509–11514 (2014).
73. Wernegreen, J. J. & Moran, N. A. Vertical transmission of biosynthetic plasmids in aphid endosymbionts (*Buchnera*). *J. Bacteriol.* **183**, 785–790 (2001).
74. Moeller, A. H. et al. Cospeciation of gut microbiota with hominids. *Science* **353**, 380–382 (2016).
75. Ochman, H. et al. Evolutionary relationships of wild hominids recapitulated by gut microbial communities. *PLoS Biol.* **8**, e1000546 (2010).
76. Guittar, J., Shade, A. & Litchman, E. Trait-based community assembly and succession of the infant gut microbiome. *Nat. Commun.* **10**, 512 (2019).
77. Vega, N. M. Experimental evolution reveals microbial traits for association with the host gut. *PLoS Biol.* **17**, e3000129 (2019).
This study uses experimental evolution of *A. veronii* in a zebrafish host system to demonstrate that improved microbial motility can be a key adaptation for successful microbial colonization of hosts.
78. Kuthyar, S., Manus, M. B. & Amato, K. R. Leveraging non-human primates for exploring the social transmission of microbes. *Curr. Opin. Microbiol.* **50**, 8–14 (2019).
79. Sarkar, A. et al. Microbial transmission in animal social networks and the social microbiome. *Nat. Ecol. Evol.* **4**, 1020–1035 (2020).
80. Parker, A., Lawson, M. A. E., Vaux, L. & Pin, C. Host-microbe interaction in the gastrointestinal tract. *Environ. Microbiol.* **20**, 2337–2353 (2018).
81. d'Hennezel, E., Abubucker, S., Murphy, L. O. & Cullen, T. W. Total lipopolysaccharide from the human gut microbiome silences Toll-like receptor signaling. *mSystems* **2**, e00046-17 (2017).
This study uses computational and experimental analyses to show that lipopolysaccharide from Bacteroidales silences TLR4 signalling for the entire gut microbiota.
82. Jeyakumar, T., Beauchemin, N. & Gros, P. Impact of the microbiome on the human genome. *Trends Parasitol.* **35**, 809–821 (2019).
83. Aldunate, M. et al. Antimicrobial and immune modulatory effects of lactic acid and short chain fatty acids produced by vaginal microbiota associated with eubiosis and bacterial vaginosis. *Front. Physiol.* **6**, 164 (2015).
84. Mackie, R. I., Sghir, A. & Gaskins, H. R. Developmental microbial ecology of the neonatal gastrointestinal tract. *Am. J. Clin. Nutr.* **69**, 1035s–1045s (1999).
85. Sprockett, D., Fukami, T. & Relman, D. A. Role of priority effects in the early-life assembly of the gut microbiota. *Nat. Rev. Gastroenterol. Hepatol.* **15**, 197–205 (2018).
86. Amato, K. R. Incorporating the gut microbiota into models of human and non-human primate ecology and evolution. *Am. J. Phys. Anthropol.* **159**, S196–S215 (2016).
87. Amato, K. R., Jeyakumar, T., Poinar, H. & Gros, P. Shifting climates, foods, and diseases: the human microbiome through evolution. *BioEssays* **41**, e1900034 (2019).
88. Kolodny, O. & Schulenburg, H. Microbiome-mediated plasticity directs host evolution along several distinct time scales. *Phil. Trans. R. Soc. B* **375**, 20190589 (2020).
89. Rudman, S. M. et al. Microbiome composition shapes rapid genomic adaptation of *Drosophila melanogaster*. *Proc. Natl Acad. Sci. USA* **116**, 20025–20032 (2019).
This study alters *D. melanogaster* microbiota by means of the diet and demonstrates that these changes altered fly body mass and population size and resulted in population genome divergence over five generations.
90. Brucker, R. M. & Bordenstein, S. R. The roles of host evolutionary relationships (genus: *Nasonia*) and development in structuring microbial communities. *Evolution* **66**, 349–362 (2012).
91. Dietrich, C., Köhler, T. & Brune, A. The cockroach origin of the termite gut microbiota: patterns in bacterial community structure reflect major evolutionary events. *Appl. Environ. Microbiol.* **80**, 2261–2269 (2014).
92. Brooks, A. W., Kohl, K. D., Brucker, R. M., van Opstal, E. J. & Bordenstein, S. R. Phylosymbiosis: relationships and functional effects of microbial communities across host evolutionary history. *PLoS Biol.* **15**, e2000225 (2016).
93. Díaz-Sánchez, S., Estrada-Peña, A., Cabezas-Cruz, A. & de la Fuente, J. Evolutionary insights into the tick hologenome. *Trends Parasitol.* **35**, 725–737 (2019).
94. Kwong, W. K. et al. Dynamic microbiome evolution in social bees. *Sci. Adv.* **3**, e1600513 (2017).
95. Tinker, K. A. & Ottesen, E. A. Phylosymbiosis across deeply diverging lineages of omnivorous cockroaches (order Blattodea). *Appl. Environ. Microbiol.* **86**, e02513–e02519 (2020).
96. Zolnik, C. P., Prill, R. J., Falco, R. C., Daniels, T. J. & Kolokotronis, S.-O. Microbiome changes through ontogeny of a tick pathogen vector. *Mol. Ecol.* **25**, 4963–4977 (2016).

97. Parker, E. S., Newton, I. L. G. & Moczek, A. P. (My microbiome) would walk 10,000 miles: maintenance and turnover of microbial communities in introduced dung beetles. *Microb. Ecol.* **80**, 435–446 (2020).
98. Novakova, E. et al. Mosquito microbiome dynamics, a background for prevalence and seasonality of West Nile virus. *Front. Microbiol.* **8**, 526 (2017).
99. Osei-Poku, J., Mbogo, C. M., Palmer, W. J. & Jiggins, F. M. Deep sequencing reveals extensive variation in the gut microbiota of wild mosquitoes from Kenya. *Mol. Ecol.* **21**, 5138–5150 (2012).
100. Colman, D. R., Toolson, E. C. & Takacs-Vesbach, C. D. Do diet and taxonomy influence insect gut bacterial communities? *Mol. Ecol.* **21**, 5124–5137 (2012).
101. Wong, A. C.-N., Chaston, J. M. & Douglas, A. E. The inconstant gut microbiota of *Drosophila* species revealed by 16S rRNA gene analysis. *ISME J.* **7**, 1922–1932 (2013).
102. Easson, C. G. & Thacker, R. W. Phylogenetic signal in the community structure of host-specific microbiomes of tropical marine sponges. *Front. Microbiol.* **5**, 532 (2014).
103. Schöttner, S. et al. Relationships between host phylogeny, host type and bacterial community diversity in cold-water coral reef sponges. *PLoS ONE* **8**, e55505 (2013).
104. Reveillaud, J. et al. Host-specificity among abundant and rare taxa in the sponge microbiome. *ISME J.* **8**, 1198–1209 (2014).
105. Webster, N. S. & Thomas, T. The sponge hologenome. *mBio* **7**, e00135-16 (2016).
106. Hentschel, U., Piel, J., Degnan, S. M. & Taylor, M. W. Genomic insights into the marine sponge microbiome. *Nat. Rev. Microbiol.* **10**, 641–654 (2012).
107. Griffiths, S. M. et al. Host genetics and geography influence microbiome composition in the sponge Ircinia campana. *J. Anim. Ecol.* **88**, 1684–1695 (2019).
108. Cleary, D. F. R. et al. The sponge microbiome within the greater coral reef microbial metacommunity. *Nat. Commun.* **10**, 1644 (2019).
109. Hentschel, U. et al. Microbial diversity of marine sponges. *Prog. Mol. Subcell. Biol.* **37**, 59–88 (2003).
110. Glasl, B., Smith, C. E., Bourne, D. G. & Webster, N. S. Exploring the diversity-stability paradigm using sponge microbial communities. *Sci. Rep.* **8**, 8425 (2018).
111. Easson, C. G., Chaves-Fonnegra, A., Thacker, R. W. & Lopez, J. V. Host population genetics and biogeography structure the microbiome of the sponge *Cliona delitrix*. *Ecol. Evol.* **10**, 2007–2020 (2020).
112. Britstein, M. et al. Sponge microbiome stability during environmental acquisition of highly specific photosymbionts. *Environ. Microbiol.* **22**, 3593–3607 (2020).
113. Nishida, A. H. & Ochman, H. Rates of gut microbiome divergence in mammals. *Mol. Ecol.* **8**, 1884–1897 (2018).
114. Sherrill-Mix, S. et al. Allometry and ecology of the bilaterian gut microbiome. *mBio* **9**, e00319-18 (2018).
115. Knowles, S. C. L., Eccles, R. M. & Baltrūnaitė, L. Species identity dominates over environment in shaping the microbiota of small mammals. *Ecol. Lett.* **22**, 826–837 (2019).
116. Kartzinel, T. R., Hsing, J. C., Musili, P. M., Brown, B. R. P. & Pringle, R. M. Covariation of diet and gut microbiome in African megafauna. *Proc. Natl Acad. Sci. USA* **116**, 23588–23593 (2019).
117. Trevelline, B. K., Sosa, J., Hartup, B. K. & Kohl, K. D. A bird's-eye view of phylosymbiosis: weak signatures of phylosymbiosis among all 15 species of cranes. *Phil. Trans. R. Soc. B* **287**, 20192988 (2020).
118. Capunitan, D. C., Johnson, O., Terrill, R. S. & Hird, S. M. Evolutionary signal in the gut microbiomes of 74 bird species from Equatorial Guinea. *Mol. Ecol.* **29**, 829–847 (2020).
119. Hird, S. M., Sánchez, C., Carstens, B. C. & Brumfield, R. T. Comparative gut microbiota of 59 neotropical bird species. *Front. Microbiol.* **6**, 1403 (2015).
120. Song, S. J. et al. Comparative analyses of vertebrate gut microbiomes reveal convergence between birds and bats. *mBio* **11**, e02901-19 (2020).
This study of 892 vertebrate species finds that the gut microbiomes of birds and bats do not follow a pattern of phylosymbiosis and that bat gut microbiomes are more similar to those of birds than to the gut microbiomes of other mammals.
121. Grond, K. et al. No evidence for phylosymbiosis in western chipmunk species. *FEMS Microbiol. Ecol.* **96**, fiz182 (2020).
122. Lutz, H. L. et al. Ecology and host identity outweigh evolutionary history in shaping the bat microbiome. *mSystems* **4**, e00511-19 (2019).
123. Gaulke, C. A., Arnold, H. K., Kembel, S. W., O'Dwyer, J. P. & Sharpton, T. J. Ecophylogenetics reveals the evolutionary associations between mammals and their gut microbiota. *mBio* **9**, e01348-18 (2017).
124. Gaulke, C. A. & Sharpton, T. J. The influence of ethnicity and geography on human gut microbiome composition. *Nat. Med.* **24**, 1495–1496 (2018).
125. Engelberts, J. P. et al. Characterization of a sponge microbiome using an integrative genome-centric approach. *ISME J.* **14**, 1100–1110 (2020).
126. Gauthier, M.-E. A., Watson, J. R. & Degnan, S. M. Draft genomes shed light on the dual bacterial symbiosis that dominates the microbiome of the coral reef sponge *Amphimedon queenslandica*. *Front. Mar. Sci.* **3**, 196 (2016).
127. Hird, S. M. Context is key: comparative biology illuminates the vertebrate microbiome. *mBio* **11**, e00153-20 (2020).
128. Youngblut, N. D. et al. Host diet and evolutionary history explain different aspects of gut microbiome diversity among vertebrate clades. *Nat. Commun.* **10**, 2200 (2019).
129. Price, J. T. et al. Characterization of the juvenile green turtle (*Chelonia mydas*) microbiome throughout an ontogenetic shift from pelagic to neritic habitats. *PLoS ONE* **12**, e0177642 (2017).
130. Cabrera-Rubio, R. et al. The human milk microbiome changes over lactation and is shaped by maternal weight and mode of delivery. *Am. J. Clin. Nutr.* **96**, 544–551 (2012).
131. Cioffi, C. C., Tavalire, H. F., Neiderhiser, J. M., Bohannan, B. & Leve, L. D. History of breastfeeding but not mode of delivery shapes the gut microbiome in childhood. *PLoS ONE* **15**, e0235223 (2020).
132. Ding, J. et al. The composition and function of pigeon milk microbiota transmitted from parent pigeons to squabs. *Front. Microbiol.* **11**, 1789 (2020).
133. Lane, A. A. et al. Household composition and the infant fecal microbiome: The INSPIRE study. *Am. J. Phys. Anthropol.* **169**, 526–539 (2019).
134. Manus, M., Kuthyar, S., Perroni-Marañón, A. G., Nuñez de la Mora, A. & Amato, K. R. Infant skin bacterial communities vary by skin site and infant age across populations in Mexico and the USA. *mSystems* https://doi.org/10.1128/mSystems.00834-20 (2020).
135. Biedermann, P. H. & Rohlfs, M. Evolutionary feedbacks between insect sociality and microbial management. *Curr. Opin. Insect Sci.* **22**, 92–100 (2017).
136. Chambers, S. A. & Townsend, S. D. Like mother, like microbe: human milk oligosaccharide mediated microbiome symbiosis. *Biochem. Soc. Trans.* **48**, 1139–1151 (2020).
This study demonstrates the importance of the microbial properties of breast milk in shaping the infant gut microbiota.
137. Donovan, S. M. et al. Host-microbe interactions in the neonatal intestine: role of human milk oligosaccharides. *Adv. Nutr.* **3**, 450S–455S (2012).
138. Gopalakrishna, K. P. & Hand, T. W. Influence of maternal milk on the neonatal intestinal microbiome. *Nutrients* **12**, 823 (2020).
139. Hasselquist, D. & Nilsson, J.-Å. Maternal transfer of antibodies in vertebrates: trans-generational effects on offspring immunity. *Phil. Trans. R. Soc. B* **364**, 51–60 (2009).
140. Flajnik, M. F. A cold-blooded view of adaptive immunity. *Nat. Rev. Immunol.* **18**, 438–453 (2018).
141. Omenetti, S. & Pizarro, T. T. The Treg/Th17 axis: a dynamic balance regulated by the gut microbiome. *Front. Immunol.* **6**, 639 (2015).
142. Chaouat, G. Reconsidering the Medawar paradigm placental viviparity existed for eons, even in vertebrates; without a "problem": why are Tregs important for preeclampsia in great apes? *J. Reprod. Immunol.* **114**, 48–57 (2016).
143. Samstein, R. M., Josefowicz, S. Z., Arvey, A., Treuting, P. M. & Rudensky, A. Y. Extrathymic generation of regulatory T cells in placental mammals mitigates maternal-fetal conflict. *Cell* **150**, 29–38 (2012).
This article argues that the adaptive immune system may have evolved in part to regulate interactions between vertebrate hosts and commensal microorganisms given the higher-diversity microbiomes observed in vertebrates.
144. Campbell, C. et al. Extrathymically generated regulatory T cells establish a niche for intestinal border-dwelling bacteria and affect physiologic metabolite balance. *Immunity* **48**, 1245–1257.e9 (2018).
This article shows that reduction of pT_{reg} cell activity alters host epithelial cells, reduces gut microbial diversity and alters gut microbiome function.
145. Yadav, M., Stephan, S. & Bluestone, J. A. Peripherally induced Tregs — role in immune homeostasis and autoimmunity. *Front. Immunol.* **4**, 232 (2013).
146. Lund, F. E. & Randall, T. D. Effector and regulatory B cells: modulators of $CD4^{+}$ T cell immunity. *Nat. Rev. Immunol.* **10**, 236–247 (2010).
147. McCoy, K. D., Burkhard, R. & Geuking, M. B. The microbiome and immune memory formation. *Immunol. Cell Biol.* **97**, 625–635 (2019).
148. Azad, M. B. et al. Gut microbiota of healthy Canadian infants: profiles by mode of delivery and infant diet at 4 months. *CMAJ* **185**, 385–394 (2013).
149. Younge, N. E., Araújo-Pérez, F., Brandon, D. & Seed, P. C. Early-life skin microbiota in hospitalized preterm and full-term infants. *Microbiome* **6**, 98 (2018).
150. Perofsky, A. C., Lewis, R. J. & Meyers, L. A. Terrestriality and bacterial transfer: a comparative study of gut microbiomes in sympatric Malagasy mammals. *ISME J.* **13**, 50–63 (2019).
151. Grieneisen, L. E. et al. Genes, geology and germs: gut microbiota across a primate hybrid zone are explained by site soil properties, not host species. *Proc. R. Soc. B* **286**, 20190431 (2019).
152. Rutayisire, E., Huang, K., Liu, Y. & Tao, F. The mode of delivery affects the diversity and colonization pattern of the gut microbiota during the first year of infants' life: a systematic review. *BMC Gastroenterol.* **16**, 86 (2016).
153. Wampach, L. et al. Colonization and succession within the human gut microbiome by archaea, bacteria, and microeukaryotes during the first year of life. *Front. Microbiol.* **8**, 738 (2017).
154. Cullender, T. C. et al. Innate and adaptive immunity interact to quench microbiome flagellar motility in the gut. *Cell Host Microbe* **14**, 571–581 (2013).
155. Suzuki, K. et al. Aberrant expansion of segmented filamentous bacteria in IgA-deficient gut. *Proc. Natl Acad. Sci. USA* **101**, 1981–1986 (2004).
156. Stagaman, K., Burns, A. R., Guillemin, K. & Bohannan, B. J. The role of adaptive immunity as an ecological filter on the gut microbiota in zebrafish. *ISME J.* **11**, 1630–1639 (2017).
157. Lochmiller, R. L. & Deerenberg, C. Trade-offs in evolutionary immunology: just what is the cost of immunity? *Oikos* **88**, 87–98 (2000).
158. Martin, L. B., Weil, Z. M. & Nelson, R. J. Seasonal changes in vertebrate immune activity: mediation by physiological trade-offs. *Phil. Trans. R. Soc. B* **363**, 321–339 (2008).
159. Ingala, M. R., Becker, D. J., Bak Holm, J., Kristiansen, K. & Simmons, N. B. Habitat fragmentation is associated with dietary shifts and microbiota variability in common vampire bats. *Ecol. Evol.* **9**, 6508–6523 (2019).
160. Phillips, C. D. et al. Microbiome analysis among bats describes influences of host phylogeny, life history, physiology and geography. *Mol. Ecol.* **21**, 2617–2627 (2012).
161. Banerjee, A. et al. Novel insights into immune systems of bats. *Front. Immunol.* **11**, 26 (2020).
162. Brun, A. et al. Morphological bases for intestinal paracellular absorption in bats and rodents. *J. Morphol.* **280**, 1359–1369 (2019).
163. Caviedes-Vidal, E. et al. Paracellular absorption: a bat breaks the mammal paradigm. *PLoS ONE* **3**, e1425 (2008).
164. Price, E. R., Brun, A., Caviedes-Vidal, E. & Karasov, W. H. Digestive adaptations of aerial lifestyles. *Physiology* **30**, 69–78 (2015).
165. Rodriguez-Peña, N., Price, E. R., Caviedes-Vidal, E., Flores-Ortiz, C. M. & Karasov, W. H. Intestinal paracellular absorption is necessary to support the sugar oxidation cascade in nectarivorous bats. *J. Exp. Biol.* **219**, 779–782 (2016).
166. Hayman, D. T. S. Bat tolerance to viral infections. *Nat. Microbiol.* **4**, 728–729 (2019).
167. O'Shea, T. J. et al. Bat flight and zoonotic viruses. *Emerg. Infect. Dis.* **20**, 741–745 (2014).
168. Zhang, G. et al. Comparative analysis of bat genomes provides insight into the evolution of flight and immunity. *Science* **339**, 456–460 (2013).
169. Ahn, M., Cui, J., Irving, A. T. & Wang, L.-F. Unique loss of the PYHIN gene family in bats amongst mammals: implications for inflammasome sensing. *Sci. Rep.* **6**, 21722 (2016).
170. Härtlova, A. et al. DNA damage primes the type I Interferon system via the cytosolic DNA sensor STING to promote anti-microbial innate immunity. *Immunity* **42**, 332–343 (2015).

171. Rathinam, V. A. K. et al. The AIM2 inflammasome is essential for host defense against cytosolic bacteria and DNA viruses. *Nat. Immunol.* **11**, 395–402 (2010).
172. Xie, J. et al. Dampened STING-dependent interferon activation in bats. *Cell Host Microbe* **23**, 297–301.e4 (2018).
173. Zhou, P. et al. Contraction of the type I IFN locus and unusual constitutive expression of *IFN-α* in bats. *Proc. Natl Acad. Sci. USA* **113**, 2696–2701 (2016).
174. Stockmaier, S., Dechmann, D. K. N., Page, R. A. & O'Mara, M. T. No fever and leucocytosis in response to a lipopolysaccharide challenge in an insectivorous bat. *Biol. Lett.* **11**, 20150576 (2015).
175. Kohl, K. D. & Dearing, M. D. Experience matters: prior exposure to plant toxins enhances diversity of gut microbes in herbivores. *Ecol. Lett.* **15**, 1008–1015 (2012).
176. Gomez, A. et al. Plasticity in the human gut microbiome defies evolutionary constraints. *mSphere* **4**, e00271-19 (2019).
177. Wooding, P. and Burton, G. *Comparative Placentation: Structures, Functions and Evolution* (Springer, 2008).
178. Andersen, K. G., Nissen, J. K. & Betz, A. G. Comparative genomics reveals key gain-of-function events in Foxp3 during regulatory T cell evolution. *Front. Immunol.* **3**, 113 (2012).
179. Mess, A. & Carter, A. M. Evolution of the placenta during the early radiation of placental mammals. *Comp. Biochem. Physiol. Part A Mol. Integr. Physiol.* **148**, 769–779 (2007).
180. Wildman, D. E. et al. Evolution of the mammalian placenta revealed by phylogenetic analysis. *Proc. Natl Acad. Sci. USA* **103**, 3203–3208 (2006).
This article concludes that the ancestral from of the mammalian placenta was the invasive, haemochorial form.

Acknowledgements
K.R.A. is supported as a fellow in CIFAR's Humans & the Microbiome programme. E.K.M. is supported by the Vanderbilt Microbiome Initiative and was partially supported by CIFAR.

Author contributions
The authors contributed equally to all aspects of the article.

Competing interests
The authors declare no competing interests.

Peer review information
Nature Reviews Microbiology thanks K. Kohl; R. Ley, who co-reviewed with N. Youngblut; and the other, anonymous, reviewer(s) for their contribution to the peer review of this work.

Publisher's note
Springer Nature remains neutral with regard to jurisdictional claims in published maps and institutional affiliations.

Supplementary information
The online version contains supplementary material available at https://doi.org/10.1038/s41579-021-00562-3.

RELATED LINKS
TimeTree: http://timetree.org